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Carbon fullerene act as potential lead molecule against prospective molecular targets of biofilm-producing multi-drug resistant *Acinetobacter baumannii* and *Pseudomonas aeruginosa*: Computational modeling and MD simulation studies

Running title: Carbon fullerene as potential lead molecule against biofilm producing *Acinetobacter baumannii* and *Pseudomonas aeruginosa*

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Abstract

This study aimed to screen putative drug targets associated with biofilm formation of multidrug-resistant *Acinetobacter baumannii* and *Pseudomonas aeruginosa* and prioritize carbon nano-fullerene as potential lead molecule by structure-based virtual screening. Based on the functional role, 36 and 83 genes that are involved in biofilm formation of *A. baumannii* and *P. aeruginosa* respectively were selected and metabolic network was computationally constructed. The genes that lack three-dimensional structures were predicted and validated. Carbon nano-fullerene selected as lead molecule and their drug-likeness and pharmacokinetics properties were computationally predicted. The binding potential of carbon nano-fullerene towards selected drug targets was modeled and compared with the binding of conventional drugs, doripenem, and polymyxin-B with their usual targets. The stabilities of four best-docked complexes were confirmed by molecular dynamic (MD) simulation. This study suggested that selected genes demonstrated relevant interactions in the constructed metabolic pathways. Carbon fullerene exhibited significant binding abilities to most of the prioritized targets in comparison with the binding of last-resort antibiotics and their usual target. The four best ligand-receptor interactions predicted by molecular docking revealed that stability throughout MD simulation. Notably, carbon fullerene exhibited profound binding with outer membrane protein (OmpA) and ribonuclease-HII (rnhB) of *A. baumannii* and 2-heptyl-4(1H)-quinolone synthase (pqsBC) and chemotaxis protein (wspA) of *P. aeruginosa*. Thus, the current study suggested that carbon fullerene probably used as potential lead molecules towards selected targets of *A. baumannii* and *P. aeruginosa* and the applied aspects probably scaled up to design promising lead molecules towards these pathogens.

Keywords: *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, Putative drug targets, Biofilm formation, Multidrug resistance, Carbon nano-fullerene, Metabolic pathway, Potential lead molecules

Introduction

With the expeditious prevalence of extreme drug resistance, *Acinetobacter baumannii* and *Pseudomonas aeruginosa* associated with nosocomial infections in immunocompromised patients declared as priority-I and priority-II pathogens respectively by WHO in 2017 (Dixit, Kumar, Kumar & Trigun, 2019; Lescat, Poirel, Tinguely & Nordmann 2019; Lin 2019; Tacconelli & Pezzani 2019). These pathogens have developed resistance to the last resort antimicrobial agents such as carbapenems and colistins affecting patients in intensive care units due to their major virulent factors such as biofilm formation and quorum sensing (Eze, Chenia, & El Zowalaty, 2019; Cilloniz, Dominedo & Torres, 2019; Garnacho-Montero & Timsit, 2019; Ibrahim, 2019; Lin 2019; Wannigama et al, 2019).

Carbapenem antibiotics are similar to penicillin, as they bind to penicillin binding proteins after they penetrate the cell wall of bacteria (Lenhard et al, 2017). Colistin interacts with lipid of the lipopolysaccharide (LPS) and destroys the bacterial membrane causing leakage of the cell content leading to cell death (Lin 2019). However, sigma factors such as *RpoN* present in *P. aeruginosa* have increased susceptibility of the organism to β -lactam based antibiotics and showed resistance to carbapenem antibiotics due to the extended-spectrum β -lactamase (Lloyd, Vossler, Nomura & Moffat 2019). Similarly, in recent years, the bacteria exhibited high adaptive resistance to colistins due to the alteration of the bacterial outer membrane through modification of LPS and reduction in its negative charge or efflux-pump over expression or over synthesis of capsule polysaccharide (Aghapour et al, 2019). On multi-pronged analysis, carbapenems and colistin were found to be associated with high fatality rate found among *A. baumannii* followed by *P. aeruginosa* associated infections (Aydin et al, 2018). Therefore, it is important to develop an alternative and promising therapeutic strategy. As biofilm production and quorum sensing are the major virulent factors in *A. baumannii* and *P. aeruginosa*, screening of biofilm producing gene products/ proteins as putative drug targets and targeting these molecular targets by emerging lead molecules provide novel therapeutic insights against multidrug resistant bacterial pathogens.

Bioinformatics and computational biology approaches are efficient methods for the rapid screening of putative drug targets and potential novel lead molecules towards the molecular

targets of biofilm producing bacterial pathogens (Malathi & Ramaiah, 2018; Lima, Del Fiol & Balcão, 2019). Carbon nano-fullerene is one of the promising nano-lead getting momentum in the field of modern drug discovery and contributes novel therapeutic intervention. (Mohajeri, Behnam & Sahebkar, 2018; Kazemzadeh & Mozafari 2019) The applications of nano-fullerene probably provide greater therapeutic potential than conventional drug candidates and curtail the emergence of multidrug resistance (Kazemzadeh & Mozafari 2019). Thus, the present study focused to identify putative drug targets by searching the biofilm producing gene products and study the interactions among these gene products in metabolic pathways. The study also prioritize the binding potential of carbon nano-fullerene against biofilm producing proteins of *P. aeruginosa* and *A. baumannii* in comparison with the binding potential of conventional antibacterials such as doripenem and polymyxin B with usual drug targets, penicillin binding protein and lipopolysaccharide binding domain respectively by computational modeling and molecular dynamic (MD) simulation studies.

Methodology

Identification and modelling of probable drug targets

Through diligent literature survey, major genes which are involved in the biofilm formation of *A. baumannii* and *P. aeruginosa* were identified. 36 and 83 genes that are involved in the biofilm formation of *A. baumannii* and *P. aeruginosa* were selected based on their functional role and involvement in the metabolic network associated with biofilm was retrieved from UniProt and KEGG (Kanehisa & Goto, 2000) (Supplementary materials, Table S1). The genes with native structures were retrieved from PDB (Kouranov et al, 2006). Among the 24 selected genes that are important in metabolic pathway related with the biofilm formation in *A. baumannii*, 21 genes that lack the 3D structures were modeled. Similarly, out of 56 genes selected from *P. aeruginosa*, 29 genes that lack the 3D structures were computationally predicted. Out of 21 selected genes in *A. baumannii*, the 3D structures of 17 genes were predicted by homology modeling and 4 genes were predicted by *ab initio* modeling by iTASSER (Zhang, 2008). Similarly, out of 29 genes from *P. aeruginosa* that lack the 3D structures, the structure of 23 genes were predicted by homology modeling and 6 genes were predicted by *ab initio* modeling. In homology modeling, the amino acid sequence of selected genes which are involved in the formation of biofilm from both the organisms that lack the 3D structures were retrieved from UniProt-KB in FASTA format and their best homologous templates were predicted by similarity

search against the non-redundant PDB using BLASTp (Altschul, Gish, Miller Myers & Lipman, 1997) as shown in **Table 1**. Clustal Omega (Sievers & Higgins, 2014; Sievers & Higgins, 2018) was used to align the template and target sequences in PIR format. 3D models were predicted by Modeller 9v20 (Webb & Sali, 2016) and RaptorX (Zhu, Wang, Bu & Xu, 2018). Superimposition, structural alignment and the analysis of root mean square deviation (RMSD) were carried out by SuperPose web server (Maiti, Van Domselaar, Zhang & Wishart, 2004). The selected target in which the homologous relationship in terms of <30% were modeled using *ab initio* method using I-TASSer server (Zhang, 2008). The descriptions of the genes modeled by *ab initio* approach are shown in Supplementary materials, Table S2.

Refinement and validation of the homology models

The energy minimization of the hypothetical models was performed on ModRefiner (Xu & Zhang, 2011). The stereo-chemical properties of the models were evaluated by QMEAN6 and ANOLEA in the SwissMODEL web server (Torsten, Jurgen, Nicolas & Manuel, 2003). The models that demonstrated residues >90% in the favored region of Ramachandran plot generated by PROCHECK (Laskowski, MacArthur, Moss & Thornton, 1993) were considered as reliable models. The quality factor pertaining to bond angles and energies was analyzed in SAVES server by ERRAT (Colovos & Yeates, 1993). The overall quality was checked by Verify 3D (Eisenberg et al, 1997). The quality of the models in comparison with experimental structures was analyzed by ProSA (Wiederstein & Sippl, 2007).

Construction of metabolic pathways

After diligent network analysis based on the function of the genes, 20 genes namely *ribA*, *ribD*, *yidC*, *ftsY*, *serS*, *dnaK*, *pgaA*, *clpP*, *recA*, *rnhB*, *lpxB*, *lpxA*, *lpxD*, *pmrA*, *pmrB*, *bfmS*, *bfmR*, *omp38*, *adeK*, *abaI* of *A. baumannii* and 54 genes such as *wspB*, *wspA*, *qseC*, *kdpE*, *pmrA*, *pmrB*, *oprJ*, *pcdQ*, *pqsB*, *pqsC*, *wspD*, *pilJ*, *pelC*, *yidC*, *ribA*, *wspE*, *lasdS*, *pelE*, *pelG*, *ribD*, *wspR*, *chpA*, *pelD*, *pilY1*, *ftsY*, *5z38*, *ladS*, *gacS*, *pelA*, *pelB*, *plcH*, *rhlB*, *lecA*, *rhlC*, *bapA*, *algC*, *phzF1*, *csrA*, *pqsh*, *pqsA*, *rhlA*, *dfa3(pqsE)*, *lasI*, *mvfR*, *phzA2*, *rhlI*, *sdiA (rhlR)*, *hfq*, *vfr*, *lasR*, *pqsD*, *qseC*, *pilJ* of *P. aeruginosa* were selected and used to construct metabolic pathways by

String database version 9.1 and Cytoscape based on their function in the formation of biofilm and interactions of the proteins were analyzed.

Pharmacokinetic studies of carbon nano-fullerene

The 3D structure of carbon nano-fullerene was retrieved from ChemSpider (Pence & Williams, 2010) (ChemSpider ID: 110185). Prediction of pharmacokinetic features of carbon nano-fullerene was carried out by admetSAR (Yang et al, 2019) and PreADMET (Veber et al, 2002) tools. Drug likeliness features of carbon nano-fullerene were analyzed on PreADMET using Lipinski's rule of five, Lead-like rule, MDDR-like rule, CMC-like rule and WDI-like rule (Lipinski, Lombardo, Dominy & Feeney 1997; Ghose, Viswanadhan, & Wendoloski, 1999; Oprea, 2000; Teague, Davis, Leeson & Oprea 1999). The ADME features were computationally predicted by PreADMET using statistical models for blood brain barrier penetration, caco2 (heterogeneous epithelial colorectal adenocarcinoma) cell permeability, plasma protein binding, HIA% (human intestinal absorption), MDCK cell permeability and skin permeability. Toxicity was predicted in terms of hERG inhibition, mutagenicity, rodent carcinogenicity and fish toxicity in Minnow and Medaka.

Molecular docking studies

Molecular docking was performed with the conventional drug such as carbapenem (Doripenem PubChem ID: 73303) and colistin-polymyxin B (Pubchem ID: 49800004) towards their known targets as they are used as the last line of drugs against *A. baumannii* and *P. aeruginosa*. Doripenem and polymyxin B were docked with the drug targets such as penicillin-binding protein (PBP) (Han et al, 2011) (PDB ID: 3UDF) and lipopolysaccharide binding domain (PDB ID: 4N4R) (Dong et al, 2014) respectively by AutoDock Vina (Trott & Olson, 2010). Similarly, the binding potential of carbon nano-fullerene towards the selected drug targets in biofilm formation of both *A. baumannii* and *P. aeruginosa* were predicted using flexible body docking by AutoDock Vina. The binding sites of selected drug targets were predicted by Depth server and AutoDock grid program. The best docked poses were screened based on the minimum binding energy (kcal/mol), cluster RMSD, interacting residues and number of interacting residues and the types of interactions. The docked complexes were visualized by MGL tools (Morris et al, 2009).

Molecular dynamic simulation

Four best docked complexes of fullerene with *wspA* and *pqsBC* of *Pseudomonas aeruginosa* and *rnhB* and *OmpA* of *Acinetobacter baumannii* were simulated by MD to confirm their stability and dynamics. Schrödinger's suite's Desmond module (Desmond Molecular Dynamics System, 2009) was utilized to perform molecular dynamic simulation. Carbon nano-fullerene and the proteins were exploited in terms of various properties for simulation. The simulations were performed at 300K using the NPT ensemble for 100ns. The trajectories of RMSD and RMSF were obtained using the following formula:

$$RMSD_x = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(r_i^t(t_x) - r_i(t_{ref}) \right)^2}$$
$$RMSF_i = \sqrt{\frac{1}{T} \sum_{t=1}^T \left\langle \left(r_i^t(t) - r_i(t_{ref}) \right)^2 \right\rangle}$$

Where t_{ref} is the reference time, N is the number of atoms, r' is the position of the selected atoms in frame x after superimposing on the reference frame, where frame x is recorded at time t_x ; T is the trajectory time over which the RMSF is calculated and r is the position of atom i in the reference at time t_{ref} ,

The RMSD, RMSF, secondary structure of proteins, protein properties, ligand torsion profile, protein - ligand contacts, properties of protein and ligand and their trajectories were analyzed.

Results and discussion

Identification of probable drug targets

The number of genes selected for this study after assiduous analysis were 83 for *P. aeruginosa* and 36 for *A. baumannii* based on their virulent traits. The major genes selected are tabulated in supplementary materials (Table S1). The genes with native structures were retrieved from PDB in which the 3D structures of 3 and 29 genes of *A. baumannii* and *P. aeruginosa* respectively were available in their native form. However, 33 and 54 genes that are important in the virulent functions of *A. baumannii* and *P. aeruginosa* are not available in their native forms. Importantly, the gene *pqsBC* present in *P. aeruginosa* is required for the biosynthesis of the quorum sensing signaling molecules which are essential for virulence and biofilm formation. The gene *wspA*

functions as a transmembrane signaling receptor involved in the chemotaxis and *pilA* that encodes for fimbrial protein forms single-species biofilm on inanimate substrate. Similarly, *ompA* present in *A. baumannii* involved in the pathogenesis and gene *rnhB* in *A. baumannii* involved in RNA catabolic processes. There are several other targets selected in the current study, for examples, *OprD*, the outer membrane protein, which is involved in the resistance against carbapenems and β -lactams (Li, Luo, Williams, Blackwell & Xie 2012). The *pmrB* gene encoding the signal sensor kinase, *PmrA/PmrB* of the two-component system increases the resistance towards polymyxins in *P. aeruginosa* (Eichenberger & Thaden, 2019; Diaz Caballero et al, 2015).

Multi-target therapeutic design involved in the identification of targets which could be targeted to achieve optimal therapeutic benefit. Computer aided screening is one such approach that is effective for the screening of novel therapeutic agents towards probable drug targets. It is reported that proteins are linked for multiple target drug design in case the ligands consist of similar structure and properties therefore have similar active sites (Batool, Kamal, Rizvi & Rashid, 2018). A major concern probably to obtain MIC values for ligand compared to the natural receptor (Bianconi et al, 2015; Zhang et al, 2019).

Construction of metabolic pathways

The metabolic pathway constructed using STRING and CytoScape in order to study the interactions among the selected genes. This was based on the structural information available and the analysis of the interaction of the genes, which foresight the impact of inhibition of a gene with the others and how this affects the biofilm pathway (**Fig. 1**). From the constructed STRING pathway of *A. baumannii*, it is clear that, the gene *bfmS* (encodes the protein bfmS) interacts with the genes such as *ompA*, *pmrA*, *pmrB*, *bfmR*, and *abaI*. The gene also interacted with *lpxA* and *lpxD*. Thus, the inhibition of *OmpA* probably prevents the interaction with *bfmS*, *pmrA*, *pmrB*, *bfmR*, and *abaI* and disrupts the functioning of the metabolic pathway. Similarly, the constructed pathway showed that *lasR* present in *P. aeruginosa* (encodes transcriptional activator protein *LasR*) interacts with *rflB*, *hfq*, *oprJ*, *sidA*, *lasB*, *rhlI*, *rhlC*, *vfr*, *ladS*, and *gacS*. Therefore, inhibiting *lasR* probably affect the expression of major genes associated in the metabolic pathway and thereby offer insight for the selection of putative targets. Taking into consideration,

the probable inhibitory function of each gene regarding its position in the pathway and their interaction with other genes in the metabolic pathways, 24 and 56 were selected as prioritized drug targets from *A. baumannii* *P. aeruginosa* respectively.

Novel drug targets can be determined based on their involvement several metabolic pathway such as amino acid biosynthesis in *A. baumannii* and *P. aeruginosa*. A study revealed that several phenazine compounds (redox active pigmented molecules), which are involved in bacterial competition and pathogenicity are present in the metabolic pathway as major genes (Das et al, 2015). The gene such as *phz1* and *phz2* are known to produce phenazine compounds in *P. aeruginosa*, which promote biofilm production (Yan et al, 2019). Regulation of these genes carried out by Pseudomonas quinolone signal mediated system (PQS) and two acyl-homoserine lactones mediated *las* and *rhl* systems that acted as potential drug targets (Lin, Cheng, Wang & Shen2018; Balasubramanian, Schneper, Kumari & Mathee, 2013). Similarly, inhibition of major genes in the biochemical pathways probably disrupts the functioning of other genes and inhibition of the metabolic pathway responsible for biofilm formation.

Hypothetical models and their computational validity

Out of 24 selected genes that are interacted each other and involved in biofilm formation by *A. baumannii*, 21 genes that lack the 3D structures were computationally predicted in which the structures of 17 genes were predicted by homology modeling. Similarly, among the 56 selected genes from *P. aeruginosa*, 23 genes were predicted by homology modeling. The secondary structures were analyzed and the models were energy minimized to get probable bioactive, minimum energy conformations. The structural and stereo-chemical validity, structural assessments, percentage of probable errors, Z-scores in comparison with experimental structures, and the energy of the local structures were assessed by various bioinformatics tools. The qualities of the hypothetical models were predicted by homology modeling as shown in **Table 2**. From the table, it is clear that all the models showed good stereo chemical quality of the protein structures by analyzing residue-by-residue geometry and the complete structure geometry, which indicated that more than 90 % of the residues located in the most favored regions of Ramachandran plot, predicted by both ProCheck. Most of the hypothetical models demonstrated

reliable Z-score in comparisons with X-ray or NMR structures predicted by ProSA server. Most of the models determined ideal compatibility of 3D atomic model with its basic amino acid residues by assessing their structural features. The statistics of non-bonded interactions among various atom types and the plot value of the error function versus residue position were found to be less by ERRAT for most of the hypothetical models. The percentages of buried atoms in terms of statistical Z-score deviation between the hypothetical model and refined PDB structures analyzed by Prove were found to be $< 1\%$ which is considered as ideal for most of the model.

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The force field parameter ANOLEA for most of the hypothetical models showed ideal potential energy within the amino acid residues in their structures. From this structural analysis, it is clear that the hypothetical models predicted were of best quality and structural accuracy that can be considered as putative targets for structure based drug discovery.

Drug likeliness and pharmacokinetic features of carbon fullerene

The drug likeliness and pharmacokinetic features of carbon fullerene predicted by various filters available in PreADMET tools is shown in Table 3. When carbon nano-fullerene was subjected to drug likeliness prediction, the molecule showed violations against Lipinski rule of five, lead like rule, CMC like rule and WDI rule. However, it qualified MDDR like rule as shown in Table 3. This showed the ability of the lead molecules to act as a potential drug needs to be carefully studied and further enhanced to improve structural features and pharmacophoric features. Similarly, the pharmacokinetic features of carbon fullerene are also predicted (**Table 3**). The compound showed low values and computationally predicted to qualify the plasma protein binding, percentage human intestinal absorption, blood brain barrier penetration, MDCK cell permeability, skin permeability and caco2 cell permeability. The fullerene molecule subjected to computational toxicity prediction where the nano- fullerene qualified the level of toxicities in the algae test, carcinoma mouse test and daphnia test predictions. It showed medium risk against hERG inhibition, showed mutagenic properties by Ames test and showed a negative value for the carcinoma rat test, as shown in **Table 3**. This showed the toxicity of the fullerene is preambled but could be lowered to be a better, less toxic compound. The carbon fullerene was further subjected to admetSAR predictions in order to obtain further pharmacokinetic features. The results are tabulated in (Supplementary material, Table S3). The computational prediction revealed that carbon fullerene qualified the test such as acute oral toxicity, blood brain barrier, caco-2, crustacea aquatic toxicity, CYP1A2 inhibition, CYP2D6 inhibition, CYP inhibitory promiscuity, eye corrosion, eye irritation, fish aquatic toxicity, honey bee toxicity, human intestinal absorption, human oral bioavailability, acute oral toxicity, OATP1B1 inhibitor, OATP1B3 inhibitor, plasma protein binding, subcellular localization, tetrahymena pyriformis and water solubility. However, the prediction showed a warning against carcinogenicity (trinary) test. The computational predication indicated the scope of carbon fullerene being used as a probable lead molecule towards the selected targets.

Interaction of fullerene with prioritized drug targets

The 56 selected proteins in *P. aeruginosa* and 24 selected proteins from *A. baumannii* that are involved in the metabolic pathway and biofilm formation were docked with carbon fullerene. The binding potential of carbon fullerene towards all the prioritized targets of *A. baumannii* and *P. aeruginosa* predicted by molecular docking studies is shown in **Table 4**. The results of docked conformations are assessed on the basis of binding energy (kcal/mol), cluster RMS, number of weak interactions and number of interacting residues. The binding potential of carbon fullerenes towards the selected proteins predicted by molecular docking studies are shown in supplementary materials **Table 4**. Among all the interaction modeling performed between carbon fullerene and prioritized targets of *A. baumannii* and *P. aeruginosa*, the best four interaction results are discussed below. When *ompA* (outer membrane protein-Omp38), one major target present in *A. baumannii*, docked with fullerene, the binding energy of the interaction estimated to be – 11.8 kcal/mol. The major interacting residues identified in the binding cavity identified to be Trp39, Val206, Leu229, Val241, Phe274, Leu275 and Trp333 (**Fig. 2a & b**). Similarly, when *rnhB* protein (ribonuclease HII) of *A. baumannii* docked with carbon fullerene, the binding energy of the docked complex estimated to be -11.7 kcal/mol. The major amino acid residues interacted with carbon fullerene was Phe60, Arg64, Trp70, Ile72, Arg74, Ser144, Arg145, Arg147, Glu148, Glu151 and Arg 154 (**Fig. 2c & d**). Among the various interactions modeling of selected targets of *P. aeruginosa* performed by molecular docking studies, the docked complex of *pqsBC*-carbon fullerene and *wspA*-carbon fullerene demonstrated the best results. When *pqsBC* (2-heptyl-4(1H)-quinolone synthase) of *P. aeruginosa* interacted with carbon fullerene, the binding energy of best docked conformation estimated to be -12.1 kcal/mol. The main amino acid residues involved in the binding sites were found to be Trp35, Trp39, Leu172, Phe173, Val206, Ala207, Phe227, Leu229, Pro271, Phe274, Leu275 and Trp333 (**Fig. 3a & b**). Similarly, when *wspA* (chemotaxis protein) of *P. aeruginosa* interacted with carbon fullerene, the binding energy of the best docked pose estimated to be -12.6 kcal/mol. The major amino acid residues interacted in the binding cavity is identified to be Val223, Leu226, Asp227, Leu240, Ala230, Val537, Lys541 and Val 542 (**Fig. 3c & d**). The cluster RMS of four docked complexes were found to be 0.00 and stabilized by various weak interactions such as vander Waals interactions and alkyl interactions. Similarly, the carbon fullerene showed significant binding potential to all the prioritized targets of *P. aeruginosa* and *A. baumannii*. (**Table 4**).

Among various docked results, the binding potential of carbon fullerene towards two other important proteins of *A. baumannii* such as bfmR protein and zur protein are shown in Supplementary material, Fig S1. Similarly, the binding potential of carbon fullerene towards five other important biofilm producing proteins of *P. aeruginosa* such as mucR, mvfR, PA4781, pelA and pqsA are shown in Supplementary material, Fig S2.

Interaction of the conventional antibiotics with their natural targets

When carbapenem- doripenem (PubChem ID: 73303) docked with their usual target, penicillin binding protein (PDB ID: 3UDF), the binding energy of the best docked pose is estimated to be -6 kcal/mol and the major amino acid residues involved in the interactions are Ala38, Pro41, Tyr62, Ile78 Thr165, His171 and Thr276. (**Fig. 4**). Similarly, when colistin polymyxin B (PubChem ID: 49800004) docked with their natural receptor lipopolysaccharide binding domain, the binding energy of polymyxin-B and lipopolysaccharide binding domain determined to be -4.9 kcal/mol.

The present study further demonstrate that the binding of carbon fullerene showed better binding potential towards most of the selected drug targets of *P. aeruginosa* and *A. baumannii* in comparison with the binding of conventional antibacterial and their usual drug targets. The binding energy of the docked complex of doripenem- penicillin binding protein estimated to be -6.00 kcal/mol (**Fig. 4**) which the found to be high in comparison with the binding of carbon-fullerene and prioritized targets of *P. aeruginosa* and *A. baumannii* as shown in Table 4. Thus, the present study clearly demonstrated that the binding potential of carbon fullerene towards selected targets in the current study is significantly higher than that of the control indicated the probable inhibitory potential of the targets.

Stability of the best docked complexes studied by MD simulation

MD simulation performed to confirm the stability of the four best complexes selected in the current study. The docked complexes of *ompA* – fullerene, *rnhB* – fullerene, *pqsBC* – fullerene and *wspA* – fullerene were simulated using Desmond module available in Schrodinger suite for

100ns and the trajectories are depicted in the **Fig. 5 – Fig. 8**. The structural features and conformational changes of carbon fullerene assessed during the MD simulation are shown in supplementary materials Fig. S4. The conformational changes in the secondary structure elements (SSE) of each of the residues in *ompA*, *rnhB*, *pqsBC* and *wspA* proteins over time during MD simulation is shown in the supplementary materials Fig. S5-Fig. S7.

The trajectories of simulated *ompA*-fullerene complex are illustrated in **Fig. 5**. From the fig. 5a, it is clear that a deviation from 2 to 3.2Å was observed in the RMSD of the complex and a deviation between 1 to 3.2Å was observed in the RMSD of fullerene in the complex in 100ns which indicated acceptable conformational changes of the protein backbone. Fig. 5b showed that the protein – ligand interactions throughout the simulation in which the specificity of the interaction plays an important role thereby indicating efficient ligand binding based on significant water bridges, hydrogen bonds and hydrophobic interactions observed throughout the simulation. Fig. 5c illustrated that the ligand root mean square fluctuation (RMSF) (>3Å) and Fig. 5d illustrated protein RMSF (1 -2.5 Å). High fluctuations of up to 4.5 Å are observed in the protein at the residues ranging from 170 to 240 whereas low fluctuations in the other region were observed. Ligand root mean square deviation (RMSD) observed directly with respect to their conformational changes during the interaction. Fig. 5e showed the trajectories of protein-ligand interactions which represent the total specific contacts of the protein with the ligand observed to be constant throughout the simulation. It is observed that the amino acids present at the functional cavity such as Trp39, Leu172, Phe 173, Val 2016, Phe227, Leu229, Phe230, Phe240, Val241, Pro242, His269, Pro271, Phe274, Leu274, Leu275 and Trp333 are involved in the interaction indicating the specificity for the docked complex as shown in molecular docking studies. Fig. 5f depicted the major stabilizing residues present in *ompA* that are interacted with fullerene during MD simulation. Thus, from the MD simulation studies, it is clear that the interaction between *ompA* and carbon fullerene is stable for 100 ns as highlighted the binding potential of carbon fullerene.

The trajectories of simulated *rnhB*-fullerene complex are illustrated in **Fig. 6**. From the Fig. 6a, it was evident that large deviation from 2.3 to 3.5Å was observed in the RMSD of the receptor-ligand complex at 100ns which indicated that the conformational changes in the backbone structure were perfectly acceptable changes. Fig. 6b showed that the protein – ligand interactions throughout the simulation in which the specificity of the interaction plays an important role and thereby indicated the efficient ligand binding based on significant water bridges, hydrogen bonds and hydrophobic interactions observed throughout the simulation. Fig. 6c illustrated that the ligand RMSF observed to be >3Å and Fig. 6d illustrated protein RMSF found to be 0.6 -3.2 Å. High fluctuations were observed in the protein at N and C terminals whereas low fluctuations observed in other region. Ligand RMSD fluctuation observed directly corresponds to their conformational changes during the protein-ligand interaction. Fig. 6e showed that trajectories of protein-ligand interactions that indicate the total specific contacts of the protein with the ligand which found to be stable throughout the simulation. It was observed that Phe60, Arg64, Trp70, Ile72, Arg74, Val143, Arg17 and Arg154 were the major residues involved in the binding and the interaction mainly stabilized by hydrophobic interactions. Fig. 6f illustrated the major amino acid residues that involved in the binding of *rnhB* and fullerene during molecular interactions. Thus, from the MD simulation, it is clear that the binding potential of carbon fullerene towards *rnhB* stable throughout the simulation and indicated the therapeutic potential of carbon fullerene.

The trajectories of simulated *pqsBC*-fullerene complex are illustrated in **Fig. 7**. Fig. 7a showed that a deviation from 1.7 to 3.1Å observed in the RMSD of the complex at 100ns which indicated perfectly acceptable conformational changes. The major forces that stabilized the binding of the ligand and receptors are the vander Waals interactions excreted by the amino acid residues such as Trp35, Phe38, Leu172, Phe 173, Phe 203, Val 206, Phe 227, Leu 229, Met 237, Val 241, Pro242, pro271, Ala272, Phe274, Leu275, Val299, Ile301 and Trp333 (Fig. 7b). The MD simulation showed that ligand RMSF was estimated to be >3Å (Fig. 7c) and protein RMSF was estimated to be 0.6 -5.4 Å (Fig. 7d). It is clear that high fluctuations of up to 5.4 Å are observed in the protein at the residues ranging from 180 to 230. Fig. 7e showed the trajectories of protein-ligand interactions representing the total specific contacts of the protein with the ligand observed throughout the simulation. Fig. 7f depicted that Trp39, Leu172, Val206, Phe227, Leu229, Met237, Pro271, Ala272 and Leu275 are major interacting residues involved in the binding of *pqsBC* and fullerene that was identified by MD simulation.

The trajectories of simulated *wspA*-fullerene complex during MD simulation are illustrated in **Fig. 8**. A deviation from 4 to 27 Å estimated as RMSD of the protein and 3 to 26 Å observed as the RMSD of fullerene in the complex in 100 ns which indicated greater conformational changes (Fig.8a). The major stabilizing forces of the interactions are hydrophobic interactions. The amino acid residues involved in interaction with the ligand were Ala 230, Leu240, Leu502, Val537, Phe540 and Val542 (Fig.8b). The ligand RMSF and protein RMSF was observed to be >6 Å and 4 - 28 Å respectively (Fig.8cd). It is clear that greater fluctuations are observed in protein at the residues 50 and 60 (Fig.8e). Fig.8f showed that Ala230 is the major amino acid that interacted with fullerene and the interaction is stabilized by hydrophobic interaction. The interaction modeling and MD simulation clearly revealed that binding potential of carbon fullerene towards the selected drug target.

There are reviews and reports suggested that several biofilm producing genes/proteins as potential drug targets. Reports suggested that biofilm proteins are the putative drug targets of *A. baumannii* (De Silva & Kumar, 2019). Reports also suggested that quorum sensing signaling molecules can act as prospective targets for drug discovery against several bacterial targets (Jiang, Chen, Yang, Yin & Yao 2019). Although several reports suggested that the scope of quorum sensing molecules and biofilm associated proteins act as drug targets, limited studies are available on the applications of biofilm producing proteins including chemotaxis protein (*wspA*) and 2-heptyl-4(1H)-quinolone synthase (*pqsBC*) of *P. aeruginosa* and outer membrane protein (*Omp38*) and ribonuclease HII (*rnhB* protein) of *A. baumannii* as potential drug targets for the binding of carbon fullerene. The present study provides insight on the application of these proteins as the therapeutic agents. Further, this study highlighted the application of carbon nano-fullerene as potential lead molecules against the prospective targets of signaling molecules and biofilm associated proteins of both *P. aeruginosa* and *A. baumannii*. Several studies revealed that carbon based nano-particles especially fullerene can be used as promising and latest therapeutic substances against several drug resistant bacterial pathogens (Zhang et al, 2019; Maleki Dizaj, Mennati, Jafari, Khezri & Adibkia, 2015) Similarly, several recent reviews highlighted that carbon based nano-material especially fullerene act as latest developments to combat ESKAPE pathogens in the era of antimicrobial resistance (Mulani, Kamble, Kumkar,

Tawre & Pardesi, 2019). In our previous studies, we have showed the application of carbon nanofullerene as potential lead molecules against probable drug targets of multidrug resistant *Salmonella typhi* (Skariyachan, Parveen & Garka 2017). The computational prediction of drug likeliness and pharmacokinetic features showed that carbon fullerene violated several theoretical perspectives that are required to consider the drug likeliness of carbon fullerene, one of the probable reasons could be the larger size of carbon fullerene. However, reviews suggested that due to greater surface area, ideal optical activity, multifunctional surface chemistry, the carbon based nano-material possess excellent biocompatibility, drug-loading capacity and lack of immunogenicity (Mohajeri, Behnam & Sahebkar 2018). Several recent reports suggested that carbon fullerenes depicted several unique structural features and possess appropriate features for interaction with several targets (Kazemzadeh & Mozafari 2019).

The present study withdrawn conclusions based on molecular interaction modeling and MD simulation studies. Thus, the prediction needs to be validated in experimental level. There are chances that the predications probably show variations in experimental analysis. Thus, the experimental assays and *in vitro* studies can be improved based on the prediction results. Further, the present study validated the stability of the best four complexes by MD simulation. Thus, the stability of all docked complex of the selected drug targets of *P. aeruginosa* and *A. baumannii* that are involved in the biofilm formation, quorum sensing and signaling molecules and their energy calculation by MMPBSA and MMGBSA need to be performed to find the potential application of carbon fullerene against these targets. Thus, the present study surely serves as an eye-opener for all the further studies.

Conclusion

Acinetobacter baumannii and *Pseudomonas aeruginosa* as priority-I and priority-II pathogen respectively by WHO and these bacteria demonstrated extreme drug resistance to most of the conventional antibacterial agents including the last resort antibiotics such as carbapemems and colistins and became a public health threat in 21st century. The mortality and morbidity rate due to the infectious caused by these bacteria are alarmingly increased and thus essential to screen alternative therapeutic remedies. Computational biology and structure based virtual screening are the recent approaches that aids to screen promising and putative molecular targets and novel lead

molecules with ideal drug likeliness and pharmacokinetics features. The current study aimed to focus the application of quorum sensing and biofilm producing genes of *A. baumannii* and *P. aeruginosa* as prospective drug targets and the potential of carbon fullerene as the promising lead molecules towards the selected drug targets. The metabolic pathways constructed by selecting 83 genes from *P. aeruginosa* and 36 genes from *A. baumannii* based on their virulent genes and proteins demonstrated that these genes possess high level of homologous relationship and demonstrated significant role in quorum sensing and biofilm formation. The current study demonstrated that carbon fullerene showed significant binding potential to most of the prioritized targets that are involved in the biofilm formation and quorum sensing pathways of both *A. baumannii* and *P. aeruginosa* in comparison with the binding of the last resort antibiotic such as doripenem and polymyxin-B with their usual targets. Importantly, carbon fullerene exhibited profound binding with most of the prioritized targets including outer membrane protein-Omp38 (OmpA) and ribonuclease HII (rnhB) of *A. baumannii* and 2-heptyl-4(1H)-quinolone synthase (pqsBC) and chemotaxis protein (wspA) present in *P. aeruginosa* predicted by molecular interaction modeling, docking studies and MD simulation. Thus, the current study suggested that these targets can be explored for novel antibacterial drug development by utilizing carbon fullerene as promising lead molecule. The approach can be scaled up and further experimental validation can be performed for future drug discovery pipelines against multidrug resistant *A. baumannii* and *P. aeruginosa*.

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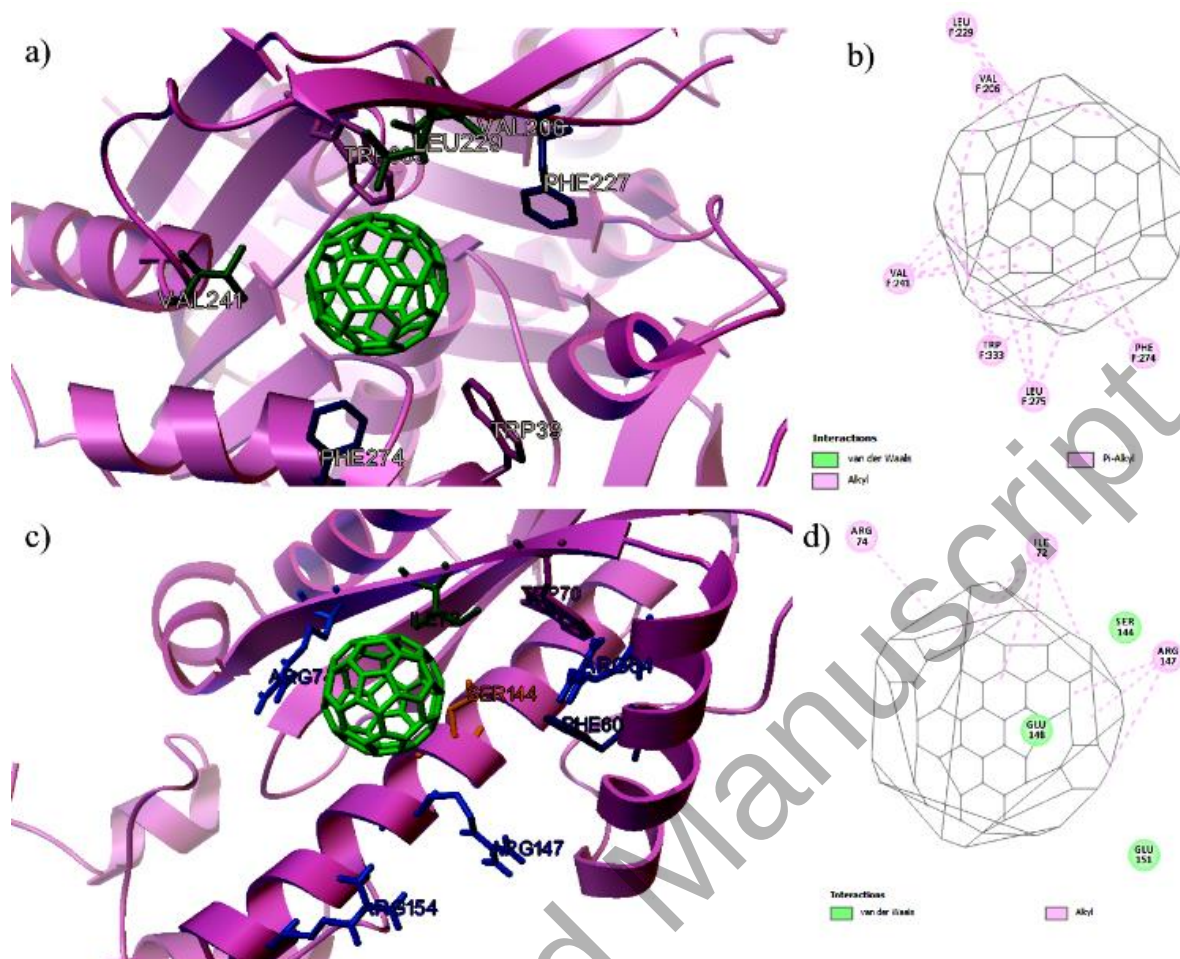


Fig. 2. Prediction of the binding potential of carbon fullerene towards outer membrane protein-Omp38 (OmpA) and ribonuclease HII (rnhB) of *A. baumannii* predicted by molecular docking by AutoDock Vina. Carbon fullerene is represented in round stick figures, the major amino acid residues are also shown in stick figures and the secondary structure present in the interacting chain is shown in ribbon structures (a) Binding site of carbon fullerene to OmpA (binding energy of -11.8 kcal/mol) protein. The major amino acid residues such as Trp39, Val206, Leu229, Val241, Phe274, Leu275 and Trp333. (b) The 2D interaction plot of carbon fullerene and OmpA protein highlighted pi-alkyl, alkyl and vander Waals interactions as the major stabilizing interactions (c) Binding site of carbon fullerene to rnhB (binding energy of -11.7 kcal/mol) protein. The major amino acid residues such as Arg64, Trp70, Ile72, Arg74, Ser144, Arg145, Arg147, Glu148, Glu151 and Arg 154. (d) The 2D interaction plot of carbon fullerene and rnhB protein highlighted alkyl and vander Waals interactions as the major stabilizing interactions

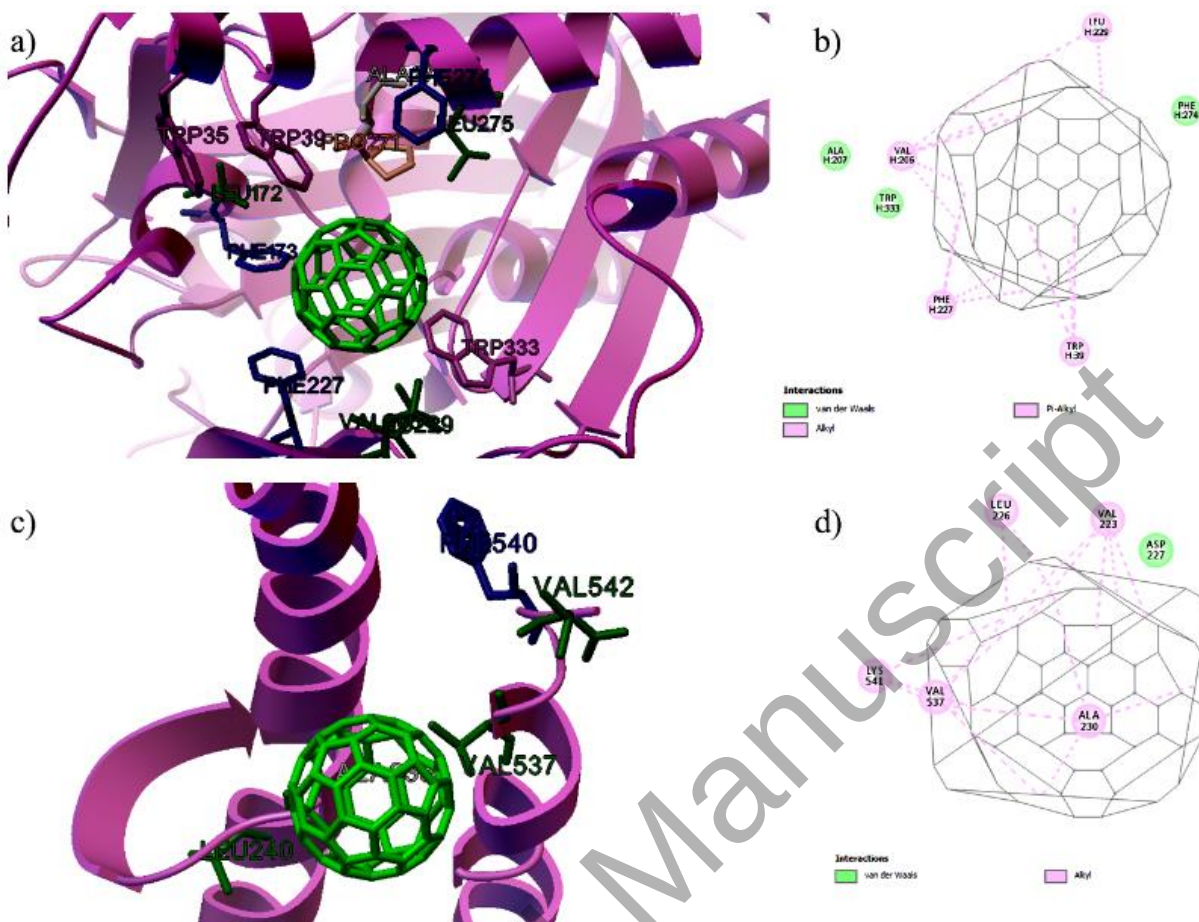


Fig. 3. Prediction of the binding potential of carbon fullerene towards 2-heptyl-4(1H)-quinolone synthase (pqsBC) and chemotaxis protein (wspA) of *P. aeruginosa* predicted by molecular docking studies by AutoDock Vina. Carbon fullerene is represented in round stick figures, the major amino acid residues are also shown in stick figures and the secondary structure present in the interacting chain is shown in ribbon structures (a) Binding site of carbon fullerene to pqsBC (binding energy of -12.1 kcal/mol) protein. The major amino acid residues such as Trp35, Trp39, Leu172, Phe173, Val206, Ala207, Phe227, Leu229, Pro271, Phe274, Leu275 and Trp333. (b) The 2D interaction plot of carbon fullerene and pqsBC protein highlighted pi-alkyl, alkyl and vander Waals interactions as the major stabilizing interactions (c) Binding site of carbon fullerene to wspA (binding energy of -12.6 kcal/mol) protein. The major amino acid residues such as Leu226, Asp227, Leu240, Ala230, Val537, Lys541 and Val 542. (d) The 2D interaction plot of carbon fullerene and wspA protein highlighted alkyl and vander Waals interactions as the major stabilizing interactions

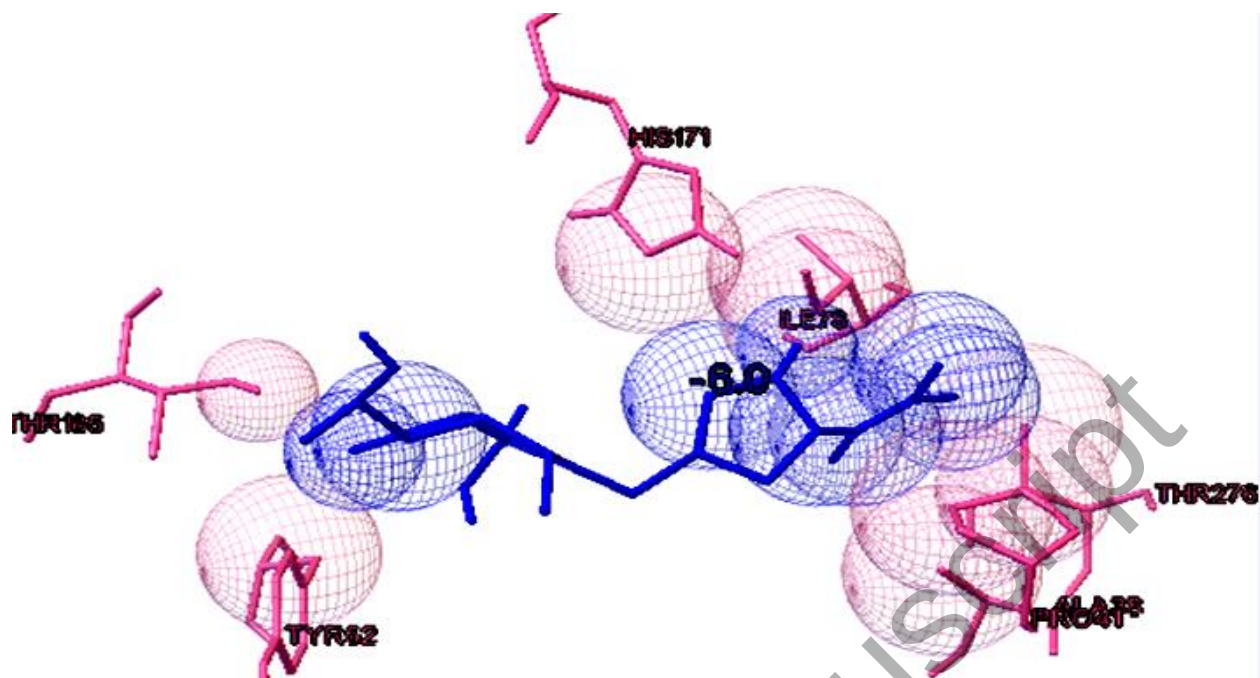


Fig. 4. The binding of one of the last resort antibiotic doripenem with its usual drug target penicillin binding protein (PDB: 3UDF) predicted by molecular docking by AutoDock Vina. The binding energy estimated to be -6 kcal/mol which is higher than the binding energy of interaction with the selected targets of *A. baumannii* and *P. aeruginosa*. The major interacting residues are identified to be Ala38, Pro41, Tyr62, Ile78 Thr165, His171 and Thr276.

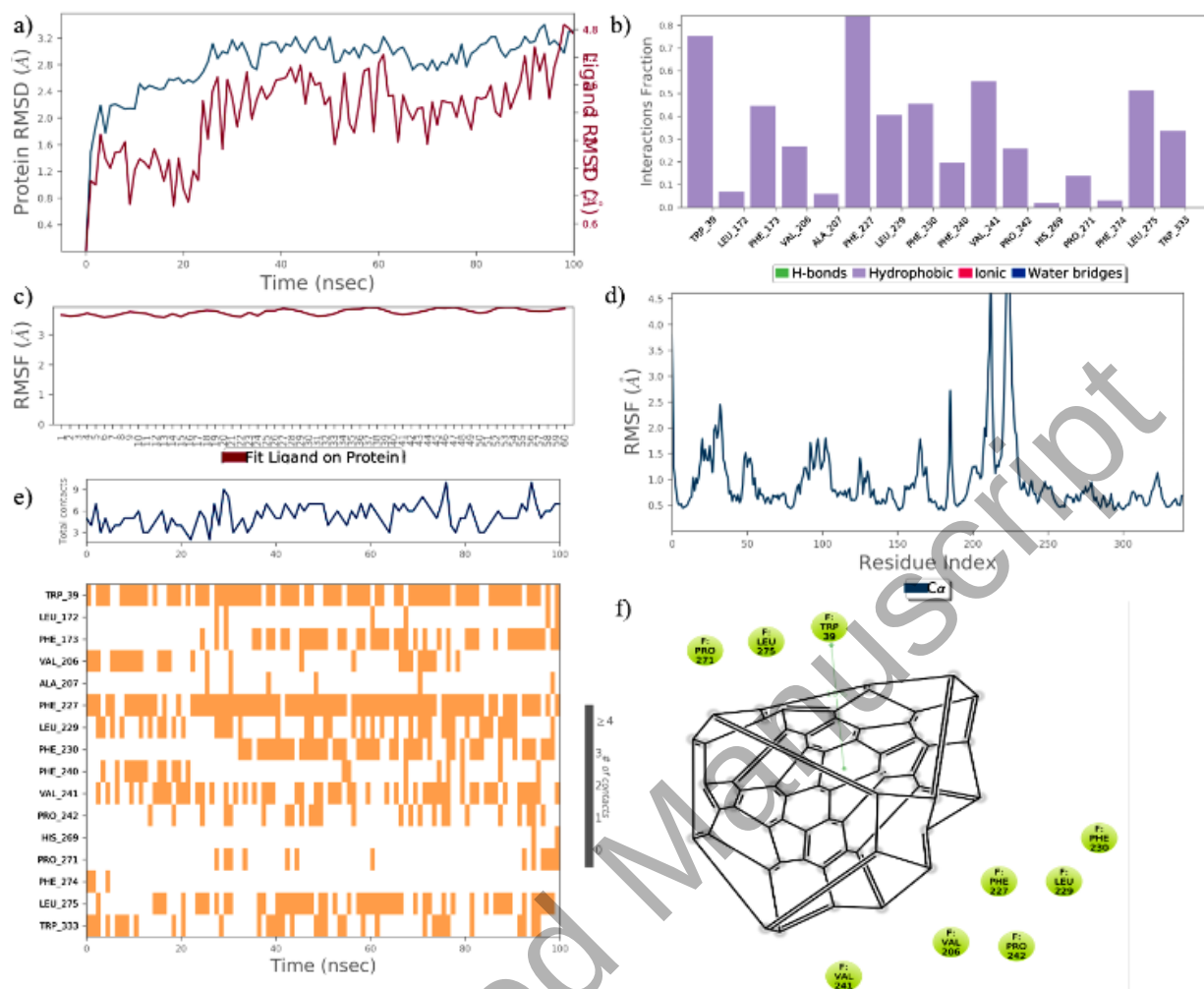


Fig. 5. The trajectories obtained by simulating ompA-fullerene complex. (a) Protein-ligand RMSD and protein RMSD (Å) on the y-axis and simulation time on x-axis (b) Histogram representing protein-ligand contacts: blue indicates water bridges, grey indicates hydrophobic interactions and green indicates hydrogen bonds (c) Ligand RMSF: RMSF (Å) on the y-axis and atom index on the x-axis (d) Protein RMSF: RMSF (Å) on the y-axis and atom index on the x-axis (e) Protein -ligand contacts: A timeline representation of interactions. Top panel shows total number of specific contacts and bottom panel shows which residues of the protein interact with the ligand and (f) The interaction between ompA and fullerene after molecular dynamic (MD) simulation. Hydrogen bonds, hydrophobic interaction and electrostatic interaction are shown in red, green, and pink, respectively. Residues shown in red, green and blue are charged, hydrophobic, and polar residues, respectively

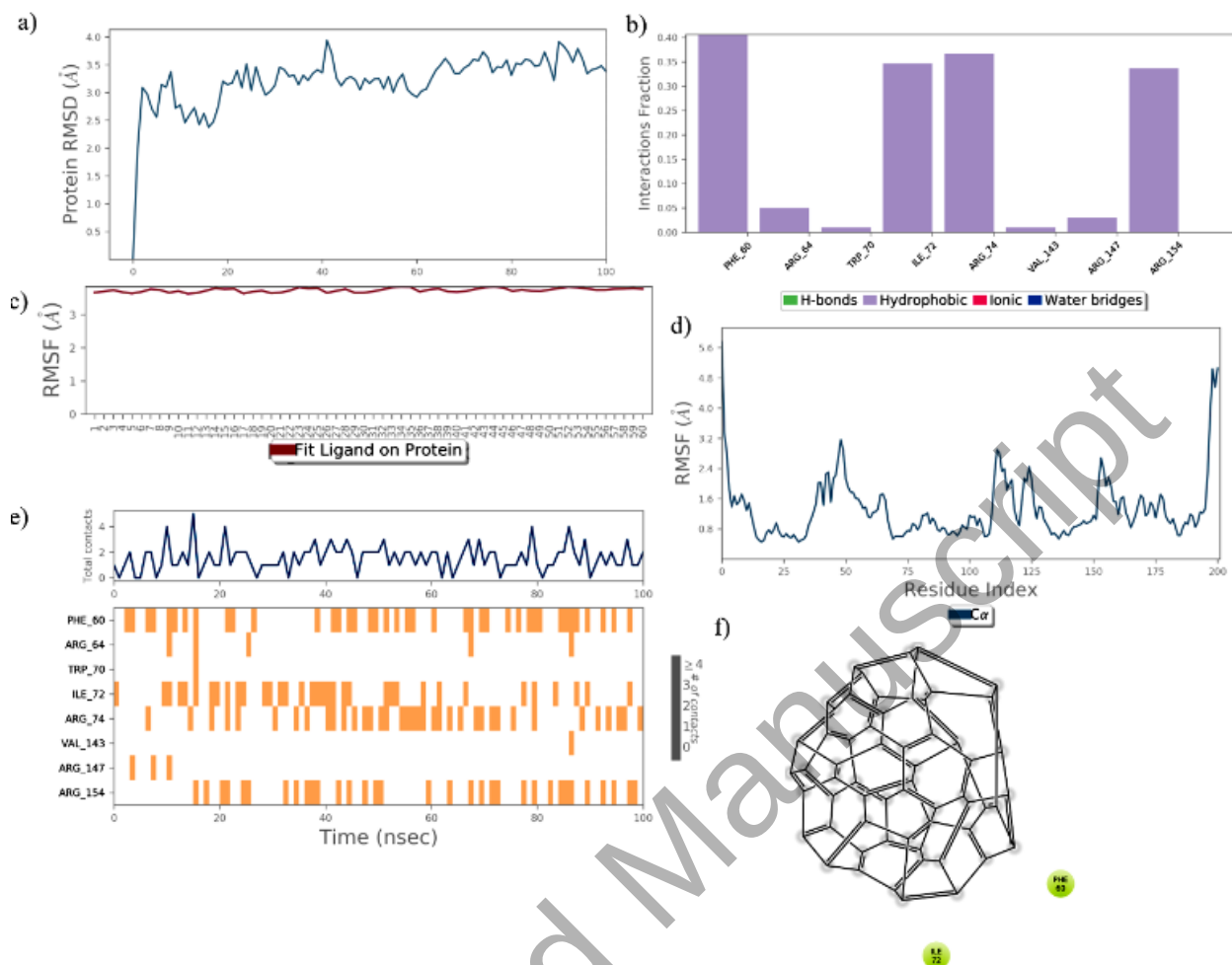


Fig. 6. The trajectories obtained by simulating rhnB-fullerene complex. (a) Protein-ligand RMSD and protein RMSD (Å) on y-axis and simulation time on x-axis (b) Histogram representing protein-ligand contacts: blue indicates water bridges, grey indicates hydrophobic interactions and green indicates hydrogen bonds (c) Ligand RMSF: RMSF (Å) on the y-axis and atom index on the x-axis (d) Protein RMSF: RMSF (Å) on the y-axis and atom index on the x-axis (e) Protein-ligand contacts: A timeline representation of interactions. Top panel shows total number of specific contacts and bottom panel shows which residues of the protein interact with the ligand and (f) The interaction between rhnB and fullerene after molecular dynamic (MD) simulation. Hydrogen bonds, hydrophobic interaction and electrostatic interaction are shown in red, green, and pink, respectively. Residues shown in red, green and blue are charged, hydrophobic, and polar residues, respectively.

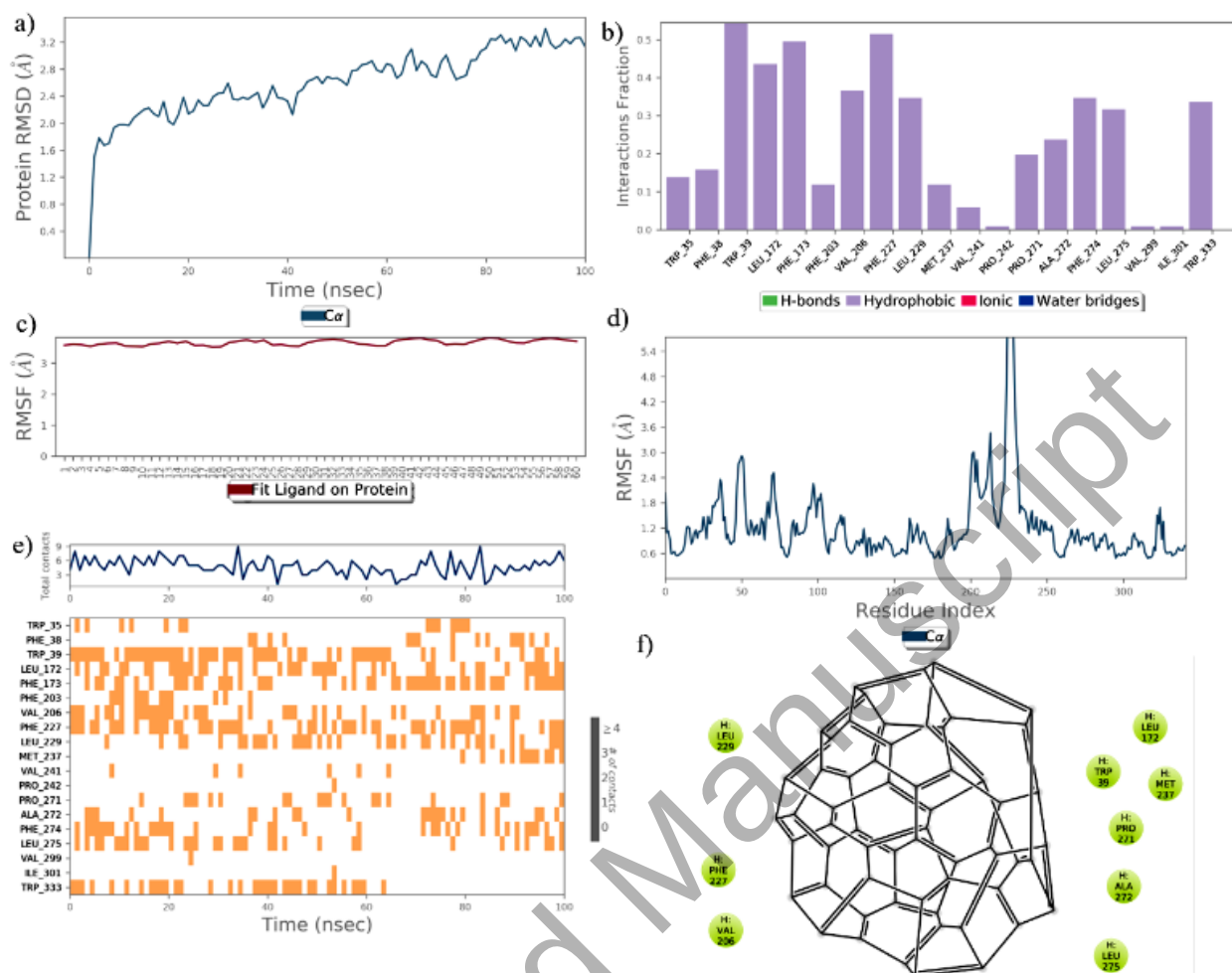


Fig. 7. The trajectories obtained by simulating pqsBC-fullerene complex. (a) Protein-ligand RMSD and Protein RMSD (Å) on the y-axis and simulation time on x-axis (b) Histogram representing protein-ligand contacts: blue indicates water bridges, grey indicates hydrophobic interactions and green indicates hydrogen bonds (c) Ligand RMSF: RMSF (Å) on the y-axis and atom index on x-axis (d) Protein RMSF: RMSF (Å) on the y-axis and atom index on the x-axis (e) Protein -ligand contacts: A timeline representation of interactions. Top panel shows total number of specific contacts and bottom panel shows which residues of the protein interact with the ligand and (f) The interaction between pqsBC and fullerene after molecular dynamic (MD) simulation. Hydrogen bonds, hydrophobic interaction and electrostatic interaction are shown in red, green, and pink, respectively. Residues shown in red, green and blue are charged, hydrophobic, and polar residues, respectively.

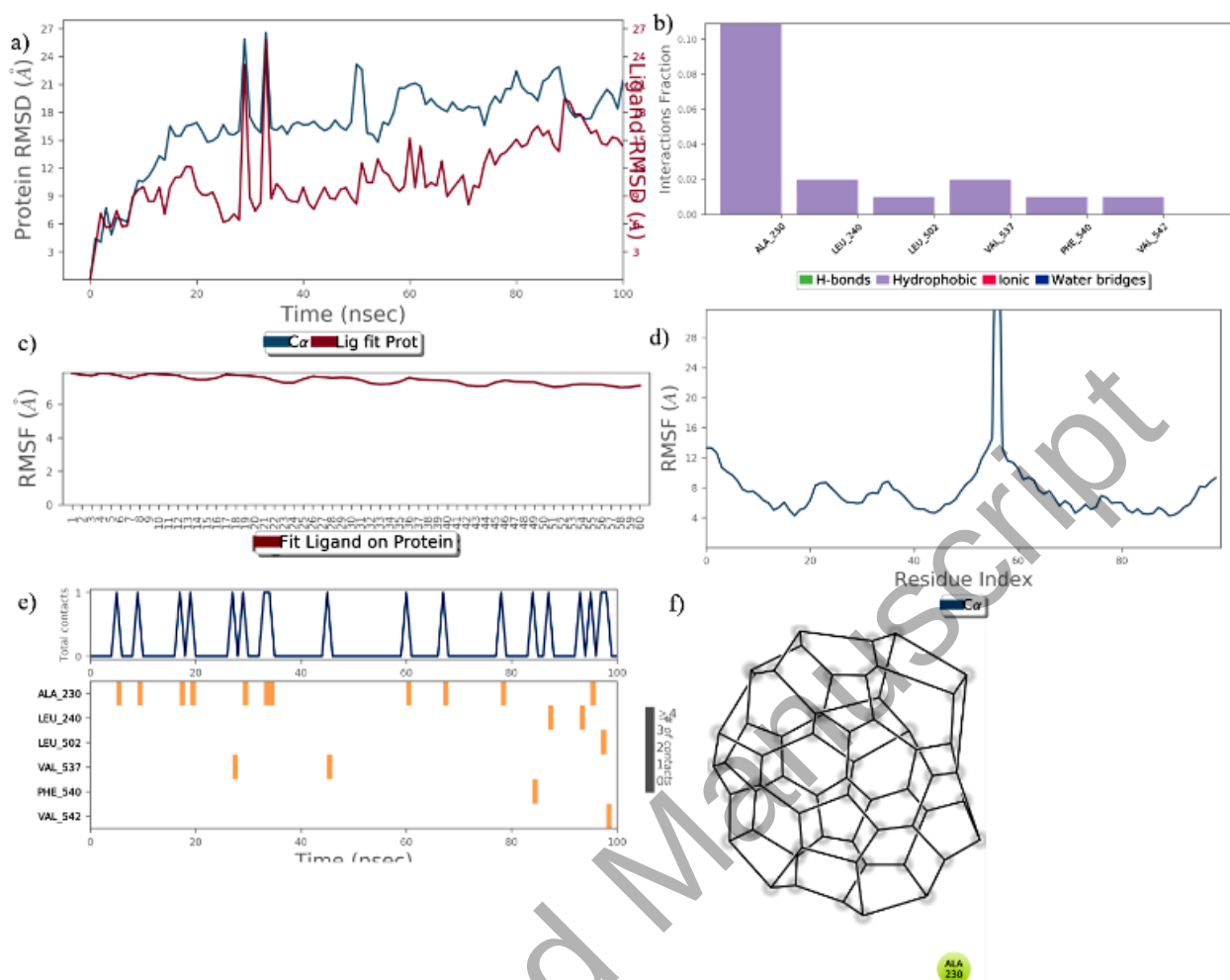


Fig. 8. The trajectories obtained by simulating wspA-fullerene complex. (a) Protein-ligand RMSD and protein RMSD (Å) on y-axis and simulation time on x-axis (b) Histogram representing protein-ligand contacts: blue indicates water bridges, grey indicates hydrophobic interactions and green indicates hydrogen bonds (c) Ligand RMSF: RMSF (Å) on the y-axis and atom index on the x-axis (d) Protein RMSF: RMSF (Å) on the y-axis and atom index on the x-axis (e) Protein-ligand contacts: A timeline representation of interactions. Top panel shows total number of specific contacts and bottom panel shows which residues of the protein interact with the ligand and (f) The interaction between wspA and fullerene after molecular dynamic (MD) simulation. Hydrogen bonds, hydrophobic interaction and electrostatic interaction are shown in red, green, and pink, respectively. Residues shown in red, green and blue are charged, hydrophobic, and polar residues, respectively.

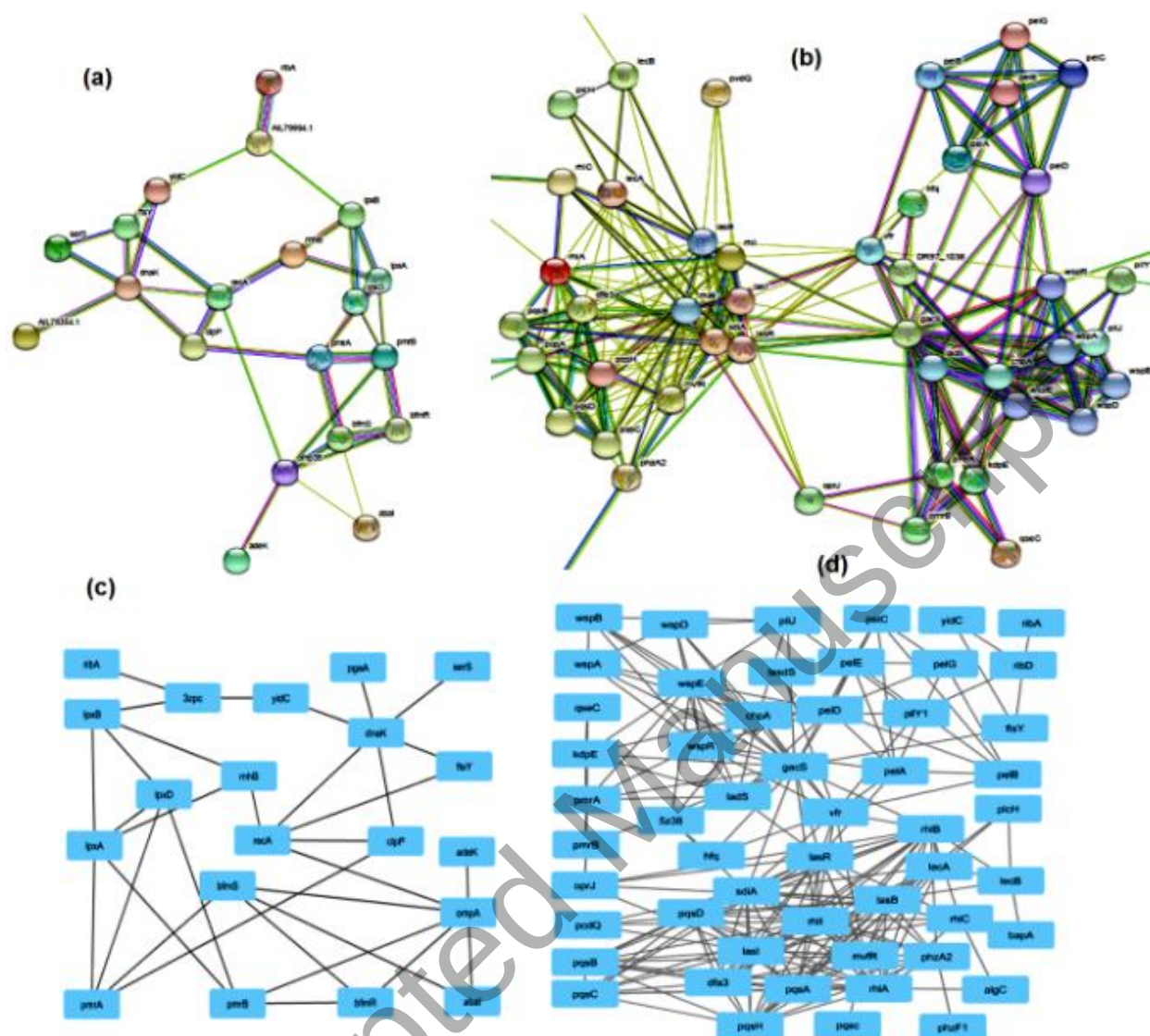


Fig. 1: The metabolic pathways indicated the genes responsible for biofilm formation constructed using STRING database for a) 24 genes of *A. baumannii* and b) 56 genes of *P. aeruginosa*. Cytoscape database gene networks created for c) *A. baumannii* and d) *P. aeruginosa*. The genes found in this pathway are involved in the formation of biofilm and the genes were selected as prospective molecular targets.

Table 1. Description of the details of the selected genes from *Acinetobacter baumannii* and *Pseudomonas aeruginosa* the features of their best homologous templates used for homology modeling

Bacteria	Gene Name	Protein Name	Sequence Length (bp)	Uniprot ID	Function	Chain	Template	Total Score (Bits)	E-value	Identity Percentage (%)	Similarity Percentage (%)	Query Coverage (%)
<i>P. aeruginosa</i>	ftsY	Signal recognition particle receptor FtsY	455	Q9I6C1	Acts as a receptor for the complex formed by SRP and the (RNC)	I	5GAD	407	2e-138	65.14	79	71
	kdpE	DNA-binding response regulator	230	A0A072ZI62	DNA binding	A	5KNY	232	4e-77	51.13	69	96
	mexH	Efflux pump membrane transporter	1,044	Q9ZH24	Involved in efflux transmembrane transporter activity	A	4DX6	1035	0.0	50.54	69	96
	mexI	Probable Resistance-Nodulation-Cell Division (RND) efflux transporter	1,029	Q9HW H4	Involved in drug transmembrane transporter activity	A	5T0O	506	2e-162	32.09	51	98
	pa1611	Histidine kinase	651	Q9I3B1	Involved in histidine phosphotrans	A	6DK8	137	3e-36	38.79	57	35

pa1750	Phospho-2-dehydro-3-deoxyheptonate aldolase	358	Q9I2Y7	fer Stereospecific condensation of phosphoenolpyruvate (PEP) and D-erythrose-4-phosphate	A	1N8F	363	6e-125	53.60	68	96
pa4497	Probable binding protein component of ABC transporter	532	Q9HVS4	Involved in peptide and dipeptide transmembrane transporter activity	A	1DPP	488	1e-168	45.17	64	95
pelC	PelC	170	Q9HZE6	Involved in single species biofilm formation	A	5T0Z	114	3e-32	39.10	58	88
phzA2	Phenazine biosynthesis protein PhzA2	162	Q9I2J9	Involved in the biosynthesis of the antibiotic phenazine	A	3FF0	228	2e-77	65.16	80	95
phzE	Phenazine-specific anthranilate synthase component I	627	G3XCK6	Involved in glutamine metabolic process and phenazine	A	3R74	696	0.0	56.49	71	99

phzF1	Trans-2,3-dihydro-3-hydroxyanthranilate isomerase	278	O69754	biosynthetic process Catalyzes the condensation of two molecules of DHHA into the phenazine ring system	A	1T6K	433	1e-154	75.18	83	100
phzG2	Probable pyridoxamine 5'-phosphate oxidase	215	Q820A5	Catalyzes the oxidation of either (PNP) or (PMP) into (PLP)	A	1T9M	436	4e-158	100	100	99
pilY1	Type IV pilus biogenesis factor PilY1	1,163	S0HPF7	Involved in pilus assembly	A	3HX6	1127	0.0	98.56	99	47
pmrA	two-component system regulator	221	Q9HVV32	Involved in DNA-binding	A	4S04	174	2e-54	43.46	62	96
ribD	Riboflavin biosynthesis protein	373	A0A0C7CY76	ATP binding	A	2O7P	364	2e-124	54.67	68	96
rhlB	ATP-dependent RNA helicase	397	Q9HXE5	Involved in RNA degradation.	A	5IVL	275	1e-88	38.89	59	95
rhlI	Acyl-homoserine-lactone synthase	201	P54291	Acts in elastase biosynthesis regulation	A	3P2F	123	5E-35	42.77	58	79
rhlR	Regulatory protein	241	P54292	Necessary for	A	4Y17	191	2e-60	40.76	60	98

ribA	GTP cyclohydrolase-2	205	Q9HW Y1	transcriptional activation of the rhlAB genes Catalyzes the conversion of GTP to DARP, formate and pyrophosphate	A	2BZ0	254	9e-87	63.2 7	74	95
trpE	Anthranilate synthase component 1	492	P2058 0	an intermediate in the biosynthesis of L-tryptophan	A	5CWA	369	1e-122	45.7 7	60	97
trpG	Anthranilate synthase component 2	201	P2057 6	an intermediate in the biosynthesis of L-tryptophan	B	1QDL	175	2e-55	44.9 5	64	95
wspA	Probable chemotaxis transducer	542	Q9HX T3	Involved in transmembrane signaling receptor activity	H	3JA6	90.9	1e-19	30.0 4	56	42
wspE	chemotaxis fusion protein	769	Q9HX T7	Involved in transferase activity	A	3JA6	228	2e-67	35.1 9	59	44
zur (np20)	Transcriptional regulator	167	A0A06 9PZQ7	Involved in DNA-	A	4MTD	93.6	3e-24	38.4 1	55	82

*Acinetobacter
baumanni*

abaI	Acyl-homoserine-lactone synthase	184	B0FLN1	binding transcription factor activity Required for normal biofilm development. Involved in efflux	A	3P2F	146	2e-44	45.61	60	92
adeK	outer membrane factor	484	Q24LT6	transmembrane transporter activity	A	1WP1	346	2e-144	44.20	62	89
cepI	Acyl-homoserine-lactose synthase	183	G0Z0A0	quorum sensing.	A	3P2F	147	1e-44	46.78	61	92
csuE	Protein CsuE	339	A0A0J1AAE3	Biofilm formation	D	6FJY	420	5e-148	63	80	92
clpP	ATP-dependent protease	201	A0A1G5F2E2	serine-type endopeptidase activity	A	4JCR	199	8e-65	55.88	79	84
dnaK	Chaperone protein	646	V5VIH3	Acts as a chaperone receptor for the complex of signal	A	2KHO	882	0.0	71.55	84	93
ftsY	Signal recognition particle receptor	368	V5VC71	recognition particle and the ribosome-nascent chain	I	5GAD	381	2e-129	60	78	90

gp45	sliding clamp, C terminal	229	A0A335Q6R3	Involved in DNA replication	A	1B77	224	4e-74	48.48	67	100
hfq	RNA-binding protein	168	V5VC31	binds with high specificity to tRNAs	A	1U1S	130	5e-40	79.22	90	45
lpxB	Lipid-A-disaccharide synthase	391	A0A059ZK65	Involved in formation of lipid A disaccharide	A	5W8S	296	1e-97	41.56	60	98
lpxD	UDP-3-o-acylglucosamine N-acyltransferase	356	V5VDY4	Involved in the biosynthesis of lipid A, a phosphorylated glycolipid	A	4E75	716	0.0	99.15	99	99
phnB (ABUW_0646)	DNA binding - methyltransferase	147	A0A0D5YF96	Involved in methyltransferase activity	A	1U6L	77.4	2e-18	31	48	99
pmrA	DNA-binding regulator	224	E2FGC2	Involved in phosphorelay signal transduction system	A	4S04	161	3e-49	39.46	60	99
qseB	Transcriptional regulator	200	A0A1G5LU56	Involved in phosphorelay signal transduction system	A	4S04	176	5e-54	41.94	61	96
recA	Protein	349	V5VD	Involved in	A	2REC	508	0.0	74.0	86	93

			Z0	catalysis of the hydrolysis of ATP in the presence of single-stranded DNA, Catalyzes the conversion of GTP to (DARP), formate and pyrophosphate						9		
ribA	GTP cyclohydrolase-2	200	V5V9J6	Specifically degrades the RNA of RNA-DNA hybrids Required complex formation of integral membrane proteins	A	2BZ0	222	4e-74	56.85	71	98	
rnhB	Ribonuclease HII	201	A0A335XJ29		A	3O3G	165	3e-51	54.89	66	91	
yidC	Membrane protein insertase	586	A0A0E1JRS7		A	4UTQ	436	3e-147	46.34	64	85	

Table 2. Structural assessment and stereo-chemical validation of the hypothetical models of the selected gene products involved in the constructed metabolic pathways of *A. baumannii* and *P. aeruginosa* predicted by homology modeling approach

Organism		Structural quality of the hypothetical models predicted by various bioinformatics tools											
	Selected gene product	RamPage					Verify3D	Errat	Prove	Procheck			Anole ^a
		Favoured Region	Allowed Region	Outlier Region	ProSa (Z-score)	Errors				Warning	Pass		
<i>Acinetobacter baumannii</i>	abaI	95.6	2.7	1.6	-2.31	66.85-F	A:80	Pass	4	2	2	-1069	
	adeK	97.1	2.3	0.6	-8.17	56.20-F	A:83.111	Pass	3	3	2	-1156	
	cepI	84.5	3.9	1.7	-2.61	75.96-F	A:60.5714	Pass	5	1	2	-916	
	clpP	94.5	3.5	2.0	0.18	26.37-F	A:89.1429	Pass	4	1	3	-700	
	csuA	94.3	4.5	1.1	-0.63	31.46-F	A:66.4384	Pass	5	0	3	-915	
	csuE	93.5	4.7	1.8	-3.05	57.85-F	A:60.6164	Pass	4	2	2	-2015	
	dnaK	94.4	4.0	1.6	-3.43	18.89-F	A:73.1667	Pass	4	2	2	-804	
	ftsY	96.7	2.7	0.5	-0.74	9.51-F	Error	Pass	4	2	2	2128	
	Hfq	99.4	0.6	0	-2.55	51.79-F	A:60.3258	Pass	3	3	2	1102	
	gp45	99.1	0.1	0	-7.76	98.25-P	A:68.3258	Pass	3	3	2	-1530	
	lpxB	94.9	2.6	2.6	0.34	12.28-F	A:73.8667	Pass	24	2	2	-913	

	lpxD	95.5	3.7	0.8	-0.3	24.44-F	A:85.0467	Pass	3	3	2	-1188
	phnA	95.5	2.7	1.8	-1.65	8.77-F	A:48.86	Pass	5	1	2	-4
	phnB	96.6	2.1	1.4	-1.75	6.12-F	A:71.73	Pass	5	1	2	-498
	pmrA	95.9	3.2	0.9	-5.22	91.07-P	A:80.5556	Pass	4	3	2	-884
	qseB	97.3	2.2	0.4	-6.39	99.56-P	A:85.6481	Pass	2	3	3	-1175
	recA	92.5	5.5	2.0	-1.97	20.63-F	A:78.5235	Pass	4	2	2	-543
	ribA	97.5	2.5	0	-2.72	56.6-F	A:78.8571	Pass	4	1	3	-761
	rnhB	97	2.5	0.5	-6.37	7264-F	A:77.20	Pass	3	2	3	-1023
	yidC	91.4	6.2	2.4	-0.38	27.3-F	A:36.3636	Pass	4	2	2	2146
	zur	97.5	2.5	0	-1.09	21.2-F	A:80.1282	Pass	3	3	2	-435
<i>Pseudomonas aeruginosa</i>	fstY	94.7	5.1	0.2	1.9	5.27-F	Error	Pass	4	2	2	3694
	kdpE	96.5	3.1	0.4	-2.57	32.17-F	A:77.3756	Pass	3	3	2	-866
	mexH	97.6	1.8	0.6	-10.82	81.61-P	A:82.9952	Pass	3	2	3	-3825
	mexI	96.0	3.5	0.5	-11.13	80.37-P	A:75.3507	Pass	3	3	2	-1650
	np20	97	3	0	-2.78	33.53-F	A:82.5263	Pass	3	2	3	-341
	pa1611	98	1.8	0.2	3.04	Error	Error	Pass	2	4	2	2380
	pa1750	94.1	3.9	2	0.23	Error	A:78.9017	Pass	4	1	3	-595
	pa4497	98.5	1.5	0	-8.87	Pass	A:74.3640	Pass	2	5	2	-2702

pelC	97.6	2.4	0	-0.07	25.00-F	A:62.1429	Pass	3	4	2	-597
phzA2	98.1	1.9	0	-4.88	89.51-P	A:77.2414	Pass	3	2	3	-1068
phzE	98.8	0.6	0.6	-5.05	79.01-F	A:91.0345	Pass	2	2	4	-1073
phzF1	92.8	5.4	1.8	-0.28	41.73-	A:66.8016	Pass	4	2	2	-937
phzG2	95.8	2.3	1.9	-1.74	53.49-F	A:69.7561	Pass	5	1	2	-848
pilY1	95.0	4.0	1.0	-0.95	19.21-F	Error	Pass	4	3	2	6072
pmrA	97.7	1.8	0.5	-6.42	95.93-P	A:98.5915	Pass	2	4	3	-1239
rhlB	97.0	2.5	0.5	-2.8	33.74-F	A:91.4163	Pass	3	3	2	-1532
rhlI	93.5	5.0	1.5	-1.33	35.32-F	A:67.9775	Pass	5	1	2	553
rhlR	95.8	4.2	0	-4.08	42.32-F	A:91.4163	Pass	2	4	2	-769
ribA	98.5	1.0	0.5	-2.01	50.73-F	A:81.5287	Pass	4	1	3	-858
ribD	93.8	4.3	1.9	-1.73	40.21-F	A:70.5382	Pass	3	4	2	-1978
trpE	91.2	5.3	3.5	-3.1	23.78-F	A:59.3361	Pass	4	3	2	-1216
trpG	95.5	3.0	1.5	-5.94	73.63-F	A:81.25	Pass	4	2	2	-1165
wspA	92.7	5.7	1.6	-1.22	31.85-F	A:69.2453	Pass	4	2	2	-623
wspE	97.8	1.8	0.4	-2.03	45.38-F	Error	Pass	3	3	2	-2181

Table 3. Drug likeliness and pharmacokinetics (ADME and Toxicity) features of carbon fullerene predicted by PreADMET server

Drug likeliness		ADME		Toxicity	
Parameter	Prediction result	Parameter	Prediction result	Parameter	Prediction result
CMC like Rule Violations	Not qualified 3	BBB	28.2401	algae_at	0.0017194
Lead like Rule Violations	Violated 2	Caco2	22.1611	Ames_test	Mutagen
MDDR like Rule Violations	Mid structure 1	HIA	100	Carcino_Mouse	Positive
Rule of Five Violations	Violated 2	MDCK	49.3884	Carcino_Rat	Negative
WDI like Rule Violations	Out of 90% cut off 10	Plasma_Protein_Binding	100	daphnia_at	2.62E-06
		Skin_permeability	-5.0254	hERG_inhibition	Medium risk

Table 4. The binding potential of Fullerene towards the probable drug targets of *A. baumannii* and *P. aeruginosa* predicted by molecular docking studies by Auto Dock Vina

Gene name	Organism	Structure	Affinity (kcal/mol)	Cluster RMSD (Å)	Interacting residues	No. of H bonds
bfmR	<i>A. baumannii</i>	PDB ID: 5HM6	-5.8	0	GLU50, LYS6	0
ompA	<i>A. baumannii</i>	PDB ID: 4G4Y	-11.7	0	TRP39, VAL206, LEU229, VAL241, PHE274, LEU275, TRP333	0
algC	<i>P. aeruginosa</i>	PDB ID: 1K2Y	-8.5	0	ALA387	0
csrA	<i>P. aeruginosa</i>	PDB ID: 1VPZ	-7.3	0	ARG50, ILE47, ILE51, GLU54	0
hfq	<i>P. aeruginosa</i>	PDB ID: 4J6X	-8.7	0	TYR165, ASP264	0
lasB	<i>P. aeruginosa</i>	PDB ID: 6FZX	-7.7	0	GLU148, TYR155	0
lasI	<i>P. aeruginosa</i>	PDB ID: 1R05	-10.2	0	GLU195, LEU193, LEU133, GLN185	0
lasR	<i>P. aeruginosa</i>	PDB ID: 4NG2	-9.3	0	GLU145, ASN141	0
lecA	<i>P. aeruginosa</i>	PDB ID: 1UOJ	-8	0	ALA1, SER121	0
lecB	<i>P. aeruginosa</i>	PDB ID: 1OUX	-10.8	0	VAL106, LEU91, GLY93, VAL77	0
mucR	<i>P. aeruginosa</i>	PDB ID: 4Y9N	-11.2	0	PRO7, TRP44	0
mvfR	<i>P. aeruginosa</i>	PDB ID: 6B8A	-11.6	0	ASP264, LYS167, TYR165, TYR185, ASP264	0
oprM	<i>P. aeruginosa</i>	PDB ID: 1WP1	-7.7	0	GLU148, TYR155	0
pa4781	<i>P. aeruginosa</i>	PDB ID: 4R8Z	-11	0	GLY269, ALA209, GLU204, ARG272	0
pelA	<i>P. aeruginosa</i>	PDB ID: 5TCB	-11.6	0	ASP103, TYR56, PRO289	0
pelD	<i>P. aeruginosa</i>	PDB ID: 4ETX	-9.9	0	HIS208, ARG205, LEU275	0

pqsA	<i>P. aeruginosa</i>	PDB ID: 5OE3	-11.3	0	GLU89, PRO111, ASP109, LYS86	0
		PDB ID: 5DWZ			TRP35, TRP39, LEU172, PHE173, VAL206, ALA207, PHE227, LEU229, PRO271, PHE274, LEU275, TRP333	
pqsBC	<i>P. aeruginosa</i>		-12.1	0	PRO78, SER 80	0
pqsD	<i>P. aeruginosa</i>	PDB ID: 3H76	-10.1	0	ARG111, ARG288	0
pqsE	<i>P. aeruginosa</i>	PDB ID: 2Q0J	-8.7	0	LEU74, HIS121, VAL160, PRO75	0
pvdQ	<i>P. aeruginosa</i>	PDB ID: 2WYB	-8.8	0	LYS401, TYR398	0
tpbB	<i>P. aeruginosa</i>	PDB ID: 4IOB	-10.2	0	LYS193, VAL203, GLY189, ARG174	0
vfr	<i>P. aeruginosa</i>	PDB ID: 2OZ6	-9.9	0	GLY307, GLN200	0
wspR	<i>P. aeruginosa</i>	PDB ID: 3BRE	-8.6	0	PHE75, THR77, PRO78, SER80	0
phnB	<i>A. baumannii</i>	Theoretical Model	-7	0	PHE60, ARG64, TRP70, ILE72, ARG74, SER144, ARG145, ARG147, GLU148, GLU151, ARG 154	0
rnhB	<i>A. baumannii</i>	Theoretical Model	-11.7	0	SER126, HIS9, GLU7, HIS6	0
zur	<i>A. baumannii</i>	Theoretical Model	-8.9	0	PRO137, LEU164, SER75, GLY126, LEU138	0
kdpE	<i>P. aeruginosa</i>	Theoretical Model	-9.4	0	ARG145, GLU151	0
mexH	<i>P. aeruginosa</i>	Theoretical Model	-11.1	0	VAL203, HIS225	0
mexI	<i>P. aeruginosa</i>	Theoretical Model	-8.2	0	ARG162	0
np20	<i>P. aeruginosa</i>	Theoretical	-4.3	0		0

phzf	<i>P. aeruginosa</i>	Theoretical Model	-10.8	0	ASP25, PHE23, HIS225	0
rhlR	<i>P. aeruginosa</i>	Theoretical Model	-10.2	0	ALA230, LYS541	0
ribA	<i>P. aeruginosa</i>	Theoretical Model	-5.8	0	TYR182, ALA184, VAL303	0
trpE	<i>P. aeruginosa</i>	Theoretical Model	-9.6	0	LYS361, ALA467, VAL303	0
wspA	<i>P. aeruginosa</i>	Theoretical Model	-12.6	0	VAL223, LEU226, ASP227, ALA230, VAL537, LYS541	0

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